UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

TEAM ASSIGNMENT #6

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

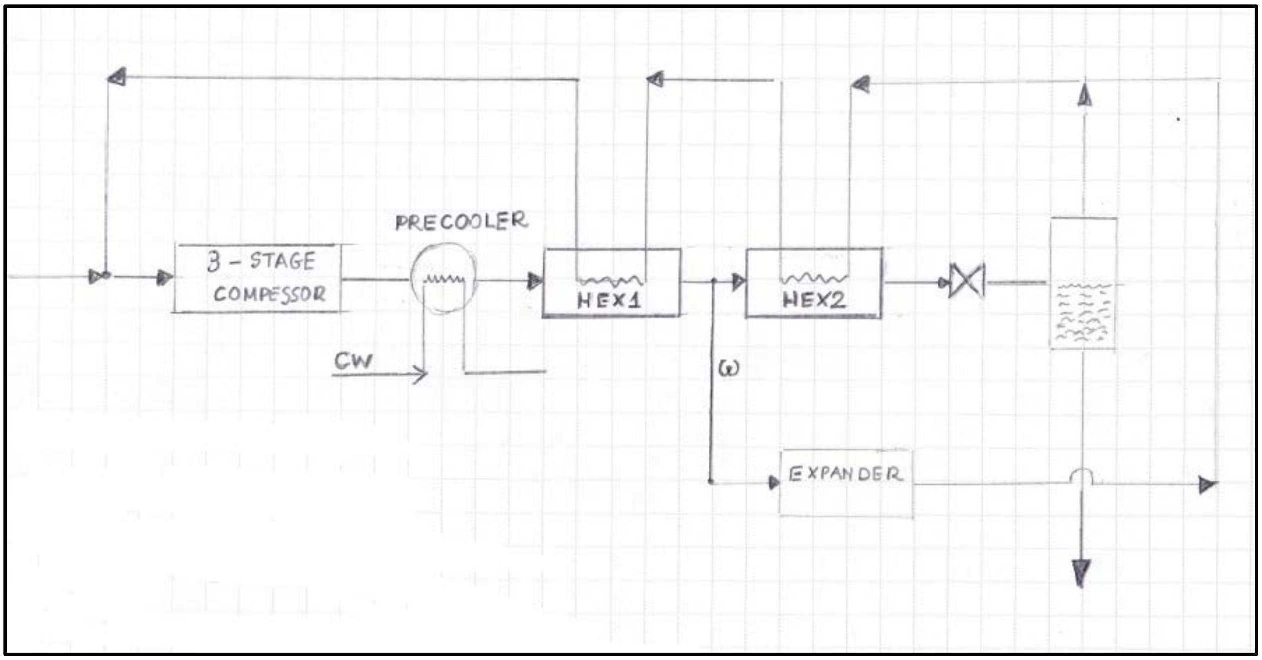
Your name:____
Due Date: Monday November 25th by 5:00pm in my office

Number of points...............................100
Your score..................................____

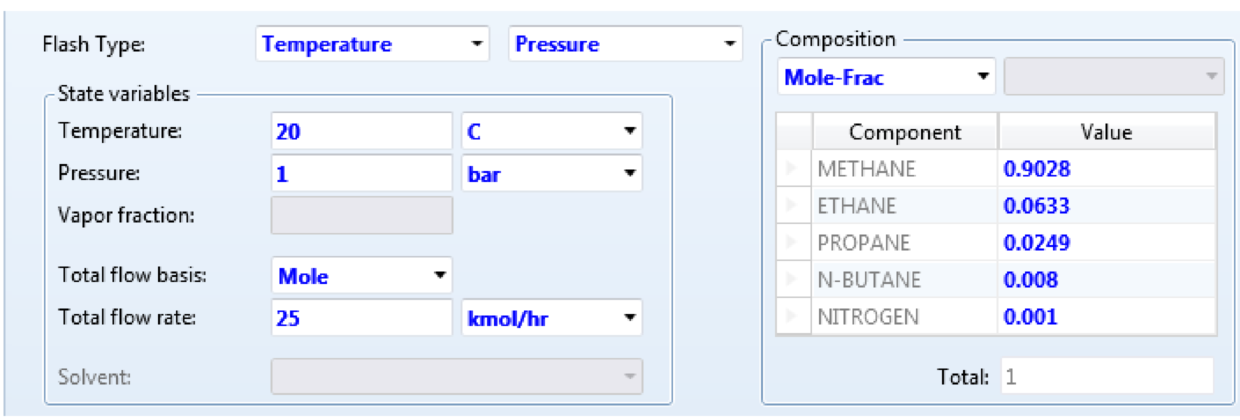
Note: All problems are worth 100 points. You final score will be the average value

Problem 6A: Liquefaction of Natural Gas. Heat Exchanger Design with ASPEN Plus EDR

Consider the liquefaction of natural gas via the Claude process shown in the process block flow diagram below: (Note: your PFD may be different than this diagram)



The flow rate, temperature, pressure and molar composition of the natural gas is given in the image below:



For the operation of this liquefaction process consider the following assumptions:

1. The efficiency of the compressor and the expander is **85%**
2. The compressor is a three stage compressor with interstage cooling. The discharge pressure of the third stage is **30bar**
3. No pressure drop in the separator. Allowable pressure drop in the pre-cooler, interstage coolers, and the heat exchangers **10psi**.

4. The flow rate of stream to the expander is **35%** of the flow of the hot stream entering the first heat exchanger (HEX1)
5. Fouling factor for all exchangers 0.002 (0.001 on each side of the heat transfer area)
6. Cooling water is available at **30 deg C**
7. The minimum temperature approach for the pre-cooler and the compressor interstage coolers is 10 deg C. The minimum temperature approach for the heat exchangers is **2 deg C**.
8. The outlet temperature of the cold stream entering the second heat exchanger (HEX2) is set by process constraints to be equal to **3 deg C**.

Prepare and Calculate:

- A detailed PFD (Process Flowsheet Diagram) with all major equipment shown and named properly:
  - Compressor: K-xxx
  - Exchangers: E-xxx
  - Expander: EX-xxx
  - Flash Tank: D-xxx
- A detailed heat and weight balance
- Detailed report of the three-stage compressor
- TEMA sheets for the two heat exchangers, the precooler and the two interstage coolers using ASPEN Plus EDR. You may need to use the **CHE433 HEX Design.xlsx** to get an estimate of the number of S&T exchangers in series depending on the value of the correction factor.
- The total capital cost of all major equipment.
- The yearly operating cost if the cost of electricity is **16.8$/GJ**, and the cost of cooling water is **0.354$/GJ.**
- If the price of the liquefied methane is **480$/1000kg** what is the annual operating profit.
- Compare the annual operating profit for the **Claude** process with that for the **Linde** process. For the **Linde** process the flow of stream to the expander is equal to zero.
- Investigate whether there is an optimal value of the flow to the expander. (Hint: Run a sensitivity analysis with respect to the split fraction to the expander and plot total cost to identify whether a minimum exists)

I will hopefully have final versions of the Capital and Operating Cost EXCEL Spreadsheets soon, because I want every single group to present the financial results with exactly the same format.